

1. Data is the rawest form of information, it contains no context and could be as simple as a number: 1

Information is the raw data with context, it gives the user additional context to the raw data, information is in essence descriptive data: 1 egg

2. Metadata gives the user more information about the data, it describes the data. Every DBMS has their advantage, depending on what the user needs the DBMS for. Most of the DBMS are free ware as well.
3. A DBMS is short for a Database Management System. This is a system that stores and manipulate the data. It also allows the user to manage the access and security to the stored data.
4. Operational databases are database that will be used daily in the operations of a company, it can hold anything from customers telephone numbers to Point of sale at a restaurant.

Analytical databases are used for analysing situation of areas, the information consist of more detailed data that allows for more descriptive and in-depth decision making, these include databases insurers use to apply a risk factor to an area.

Although there are differences in the Operational and Analytical databases, the main thing they have in common is the structure they possess. All types of databases have a structure, field, datatypes, etc. Another thing they have in common is that they are stored on hardware and uses a type of DBMS to run the database.

5. There are 4 types of NoSQL:
 - 5.1. Document Databases
 - 5.2. Key-value Databases
 - 5.3. Wide-column data stores
 - 5.4. Graph databases

Document Databases are databases consisting of attribute in one table. For example, a user entry at a library: the entry consists of the person's Name, Surname, ID, Age, Date of birth, Address, etc. All this data is linked to the Library ID in one table.

Key-value databases are datasets consisting of 2 columns: a key and an attribute. A good example of this is a list of car registration numbers (number plates) where the

key is the end/beginning of the registration number, and the attribute will be where the motor vehicle was register, for example CAA is Cape Town, GP is Gauteng and MP is Mpumalanga.

Wide-column stores are dataset for example car manufacturing per vehicle type. Meaning there is a main field (key) and multiple other columns associated with the key value. In my example the key will be the type of car, Hilux, and the columns associated with this will be the amount of bakkies built in the manufacturing process per month in the factory. This can then also assist to calculate the yearly built count.

Graph databases can be used by Insurers to calculate the risk area of where a person stays, the data are stored in nodes that are associated with two attributing factors, in the example, it would be house break-ins and proximity to busy roads for example. The two factors are the axis of the graph while the position of the node will reflect the factors.

6. SQLite does not need a sever connection of configuration. This allows for it to be able to handle small projects or data. In my line of work I deal with SQLite data sets quite often, I receive spatial data (shapefiles) in SQLite format, this databases store both text tables and Geodesic datasets. This allows for easy transport and sharing of the data. Since this is only one use, it also shows how easily SQLite can be configured to the needs of a user.
7. To explain ACID I will be using my work as an example. I work for a company that creates and maintains Local municipalities Asset Register. Currently our Software stores the Asset Register information in the DBMS PostgreSQL.

7.1. Atomicity

The hierarchy of the Asset registers are coded values, however there are other tables, reference tables that has the code and attribute calculated, these reference tables have to be joined to the register table to make sense of the register table. This means that the tables are broken up into smaller tables to make working with it easier.

7.2. Consistency

There are numerous checks run on the fields in the asset register table, checks such as minus values in financial fields where it cannot be minus values. Other checks include coded values that link to reference table (lookup tables) and will return an error if the coded value is not found in the lookup table.

7.3. Isolation

There are multiple calculated fields in the register that is calculated automatically that happens when an entry is added. Some of these fields are like the EUL, RUL and Depreciation. This falls under the Isolation of ACID.

7.4. Durability

Some of our clients host their own servers, and numerous data crashes has happen, luckily we have been able to recover and restore the asset register database every time.